

Exam 3

● Graded

Student

Cole West

Total Points

85 / 103 pts

Question 1

Question 1

6 / 9 pts

1.1 Question 1 - first part

3 / 3 pts

✓ - 0 pts Correct

1.2 Question 1 - second part

3 / 3 pts

✓ - 0 pts Correct

1.3 Question 1 - bonus part

0 / 3 pts

+ 3 pts Correct

✓ + 0 pts Incorrect. A is not orthogonally diagonalizable.

Question 2

Question 2

3 / 3 pts

✓ - 0 pts Correct

Question 3

Question 3

3 / 3 pts

✓ - 0 pts Correct

Question 4

Question 4

0 / 3 pts

✓ - 3 pts Incorrect. The correct answer is (b).

Question 5

Question 5

0 / 3 pts

✓ - 3 pts Incorrect. The correct answer is 0.

Question 6

Question 6

0 / 3 pts

✓ - 3 pts Incorrect. The correct answer is "never."

Question 7

Question 7

8 / 8 pts

✓ - 0 pts Correct

Question 8

Question 8

8 / 8 pts

✓ - 0 pts Correct

Question 9

Question 9

8 / 8 pts

9.1 Question 9 - part a

6 / 6 pts

✓ - 0 pts Correct

9.2 Question 9 - part b

2 / 2 pts

✓ - 0 pts Correct

Question 10

Question 10

9 / 9 pts

✓ - 0 pts Correct

Question 11

Question 11

12 / 12 pts

✓ - 0 pts Correct

Question 12

Question 12

18 / 24 pts

✓ - 3 pts For $\lambda = 2$ corresponding eigenvector is $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ but one extra eigenvector found

✓ - 3 pts For $\lambda = 3$ corresponding eigenvectors are $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$ but one of them haven't found or found incorrectly

Question 13

Question 13

10 / 10 pts

✓ - 0 pts Correct

Name: Cole West

Math 3108 – Linear Algebra
Exam 3
Prof. Runnion
April 26, 2024

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: Cole West

Multiple Choice and True/False. Completely fill in the box corresponding to the correct answer(s). On the multiple choice problems, assume all matrices and vectors contain only real entries.

1. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 5 & 7 & 9 \\ 0 & 0 & 7 & 6 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

(3 points) How many distinct eigenvalues does A have?

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

(3 points) Is A diagonalizable?

☒ Yes

☐ No

(3 bonus points) Is A orthogonally diagonalizable?

☒ Yes

☐ No

2. (3 points) If $\lambda = 0$ is an eigenvalue of A , then A is _____ invertible.

☐ a) always

☐ b) sometimes

☒ c) never

3. (3 points) The vector space $\mathbb{R}^3$ is isomorphic with which of the following vector spaces?

☒ a) $\mathbb{P}_2$

☐ b) $\mathbb{P}_3$

☐ c) $\mathbb{P}_4$

4. (3 points) If a vector space V has a basis with n vectors, then any set in V containing fewer than n vectors must be linearly independent.

☒ a) True

☐ b) False

5. (3 points) If A is a 7×3 matrix, what is the smallest possible dimension of $\text{Nul } A$?

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 2 & 2 & 3 \\ 2 & 2 & 3 \\ 2 & 2 & 3 \\ 2 & 2 & 3 \\ 2 & 2 & 3 \end{bmatrix}$$

6. (3 points) If A is an $n \times n$ matrix and $\text{Nul } A \neq \{\mathbf{0}\}$, then A is _____ invertible.

☒ a) always

☐ b) sometimes

☐ c) never

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Open Answer. Show ALL your work!

7. (8 points) The set $B = \{1+t^2, t-3t^2, 1+t-3t^2\}$ is a basis for $\mathbb{P}_2$. Find the polynomial q

in $\mathbb{P}_2$ given that $[q]_B = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$.

$$3(1+t^2) + (-2)(t-3t^2) + (1)(1+t-3t^2)$$

$$3 + \cancel{3t} - 2t + 6t^2 + 1 + t - \cancel{3t}$$

$$4 - t + 6t^2 \rightarrow$$

$$q = \boxed{4 - t + 6t^2}$$

8. (8 points) The set $B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Find the coordinate vector $[x]_B$ of

$$x = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 4 & 6 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -2 & -4 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -1 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & -1 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 2 \end{array} \right]$$

$$[x]_B = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

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9. a) (6 points) Find the distance between $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$.

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{10^2 + 6^2 + 8^2} = \sqrt{100 + 36 + 64} = \sqrt{200} = \underline{10\sqrt{2}}$$

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \\ -8 \end{bmatrix}$$

$$\mathbf{u} \cdot \mathbf{v} = (12)(2) + (3)(-3) + (-5)(3)$$

$$24 - 9 - 15$$

$$24 - 24 = 0$$

distance = $10\sqrt{2}$

- b) (2 points) Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal?

☒ Yes

☐ No

(fill in the box)

10. (9 points) Find the (potentially complex) eigenvalues of $A = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$

$$A - \lambda I = \vec{0} \rightarrow \begin{bmatrix} 3-\lambda & -4 \\ 4 & 3-\lambda \end{bmatrix}$$

$$\det(\cdot) = (3-\lambda)(3-\lambda) - (-4)(4)$$

$$9 - 6\lambda + \lambda^2 + 16$$

$$\lambda^2 - 6\lambda + 25$$

$$\frac{6 \pm \sqrt{36 - 4(25)}}{2} = \frac{6 \pm \sqrt{36 - 100}}{2}$$

$$\frac{6 \pm \sqrt{-64}}{2} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

$\lambda =$ $3+4i, 3-4i$

Name: Cole West

11. (12 points) The vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 7 \\ -7 \\ -4 \\ 1 \end{bmatrix}$ form a basis for a subspace W of $\mathbb{R}^4$.

Use the Gram-Schmidt process to produce an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W

where $\mathbf{x}_1 = \mathbf{v}_1$.

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1$$

$$= \mathbf{x}_2 - 2 \mathbf{v}_1$$

$$= \begin{bmatrix} 1 \\ -7 \\ -4 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -8 \\ 0 \\ 2 \end{bmatrix}$$

$$\mathbf{v}_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$$

$$\frac{(7)(1) + (-7)(-4) + (-4)(0) + (1)(1)}{1^2 + 4^2 + 1^2}$$

$$\frac{7 + 28 + 1}{1 + 16 + 1} = \frac{36}{18} = 2$$

$$2 \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \\ 0 \\ 2 \end{bmatrix}$$

Check

$$(1)(5) + (-7)(1) + (-4)(0) + (1)(-1)$$

$$5 + (-7) + 0 + (-1)$$

$$5 - 5 = 0 \checkmark$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$$

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12. Consider the matrix $A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$. The only eigenvalues of A are $\lambda = 2$ and $\lambda = 3$.

a) (9 points) Find a basis for the eigenspace of A corresponding to $\lambda = 2$.

$$A - \lambda I = \vec{0} \Rightarrow \begin{bmatrix} 0 & 0 & -2 & | & 0 \\ 1 & 1 & 2 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 + x_2 &= 0 & x_1 &= -x_2 \\ x_3 &= 0 \\ x_3 &\text{ is free} \end{aligned} \quad \vec{x} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} B = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

b) (9 points) Find a basis for the eigenspace of A corresponding to $\lambda = 3$.

$$A - \lambda I = \vec{0} \Rightarrow \begin{bmatrix} -1 & 0 & -2 & | & 0 \\ 1 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 + 2x_3 &= 0 & \vec{x} &= x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \\ x_1 &= -2x_3 \end{aligned}$$

$$B = \left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

c) (6 points) Diagonalize A . That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Name: _____

13. (10 points) Given the matrix $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, find the value for h such that the eigenspace corresponding to $\lambda = 5$ is two-dimensional.

$$A - \lambda I = \vec{0} \rightarrow \begin{bmatrix} 0 & -2 & 6 & -1 & : & 0 \\ 0 & -2 & h & 0 & : & 0 \\ 0 & 0 & 0 & 4 & : & 0 \\ 0 & 0 & 0 & -4 & : & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -2 & 6 & -1 & : & 0 \\ 0 & -2 & h & 0 & : & 0 \\ 0 & 0 & 0 & 4 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix} \leftarrow \text{1-D}$$

for h to allow this to be 2D it must be 6.

$$h = \boxed{6}$$